```
Metadata:  corpus specific: 0, document level (indexed): 0
Content:  documents: 5
```

The above method is to be used when you want to perform text processing on data present on your local machine in a folder, but if you want to load a dataset from an online repository, R allows you to directly load the dataset into your project as mentioned below.

To understand how to load data in R directly from the Internet, we use the famous Iris dataset (stored in CSV format) available from the UCI Machine Learning Repository (*http://archive.ics.uci.edu/ml/*), which provides free access to several datasets that can be used to test many machine learning algorithms.

Download and load the library RCurl which allows fetching of URIs, HTTP requests, *get* and *post* forms, and then process the results returned from the Web server, as follows:

```
library(RCurl)
url <-'https://archive.ics.uci.edu/ml/machine-learning-
databases/iris/iris.data'
 docs <- getURL(url, ssl.verifypeer=FALSE)
connection <- textConnection(docs)
dataset <- read.csv(connection, header=FALSE)
head(dataset)
   V1  V2  V3  V4          V5
1 5.1 3.5 1.4 0.2 Iris-setosa
2 4.9 3.0 1.4 0.2 Iris-setosa
3 4.7 3.2 1.3 0.2 Iris-setosa
4 4.6 3.1 1.5 0.2 Iris-setosa
5 5.0 3.6 1.4 0.2 Iris-setosa
6 5.4 3.9 1.7 0.4 Iris-setosa
```

Preprocessing

Preprocessing the text means removing punctuations, numbers, common words, etc. This step is performed to clean the data to increase the efficiency and robustness of our results by removing words that don't necessarily contribute to our analysis.

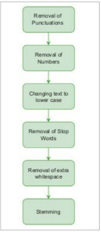

Figure 1: Text preprocessing

Removal of punctuation: To remove all punctuation, use the following command:

```
docs <- tm_map(docs, removePunctuation)
```

Removal of numbers: The following command will remove the numbers in the text:

```
docs <- tm_map(docs, removeNumbers)
```

Changing the text to lower case: The command shown below will change the text being mined into lower case:

```
docs <- tm_map(docs, tolower)
```

Removal of stop words: Common words like 'but', 'and', etc, are called stop words. The following command will remove them:

```
docs <- tm_map(docs, removeWords, stopwords("english"))
```

Removal of extra whitespaces: The following command will remove the extra whitespaces:

```
docs <- tm_map(docs, stripWhitespace)
```

Stemming: Stemming is the procedure of converting words to their base or root form. For example, *playing, played, plays*, etc, will be converted to *play.*

To perform stemming, use the command shown below:

```
library(SnowballC)
docs <- tm_map(docs, stemDocument)
```

Document term matrix

A document term matrix (DTM) depicts the frequency of a word in its respective document. In this matrix, the rows are represented by documents and the columns are represented by words. How often a word occurs in a document, is what determines the value ascribed to it—if it appears only once, its value is 1; if it appears twice, its value is 2; and if it does not appear at all, then the value is 0.

To create a DTM for our corpus, type:

```
dtm<- DocumentTermMatrix(docs)
dtm
<<DocumentTermMatrix (documents: 5, terms: 2566)>>
Non-/sparse entries: 3449/9381
Sparsity           : 73%
Maximal term length: 23
Weighting          : term frequency (tf)
```

The summary of our DTM states that there are five documents and 2566 unique words. Hence, let's give a matrix with the dimensions of 5x2566, in which 73 per cent of the rows have value 0.

To get the total frequency of words in the whole corpus, we can sum the values in a row, as follows:

```
freq <- colSums(as.matrix(dtm))
```

For that, we have to first transform the DTM into a matrix, and then sum up the rows to give a single value for each column.

Next, we sort this in descending order to get to know the terms with the highest frequency, as follows:

```
ord <- order(freq,decreasing=TRUE)
freqn[head(ord)]
```